

DR ESLAM AHMED
eliteigcse.com

CHAPTER 1 • NUMBER • TOPIC 13

Ratio Problem Solving

The longer ratio stories: three-way shares, ratio with fractions and percentages, joined ratios, before-and-after and ratio with letters.



$$\frac{3}{10} = 0.3 =$$

30%**5**

VARIANTS

5

PAST-PAPER PARTS

38

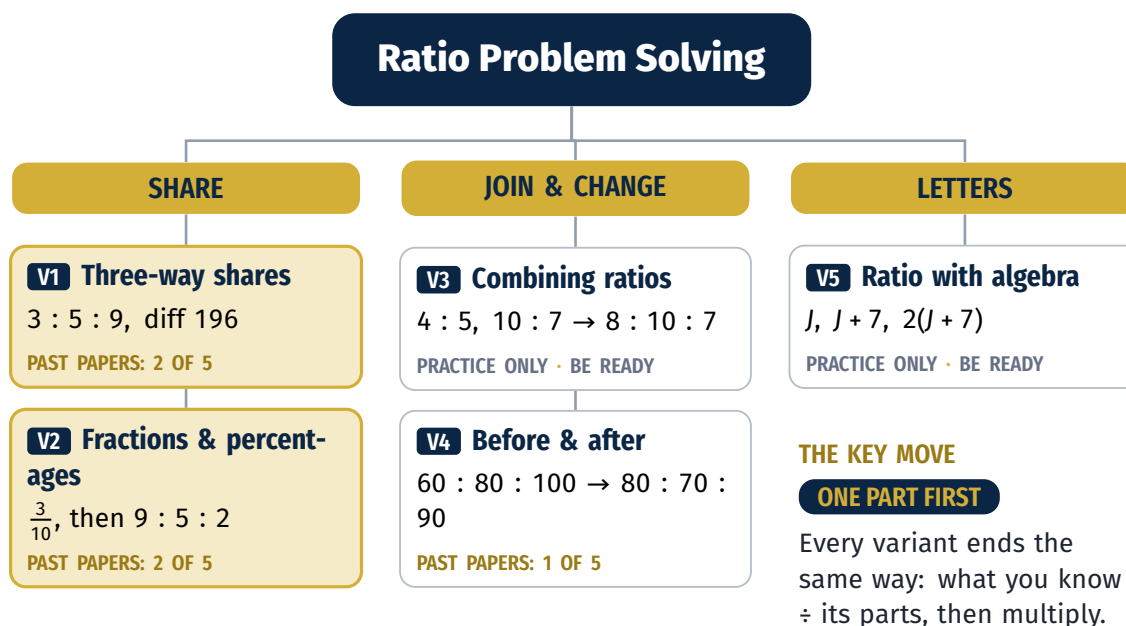
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Three-way shares**V2** Fractions & percentages**V3** Combining ratios**V4** Before & after**V5** Ratio with algebra

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Ratio problems at a glance



Your notes


★ **How often does each variant come up?**

5 past-paper questions (Jan 2020 $\rightarrow$ Nov 2025); 38 practice parts — grouped by type. Basic sharing and linking are in Topic 12.

	PAST PAPERS (OUT OF 5)		PRACTICE PARTS	
V1 Three-way shares	● ● ● ● ●	2	■ 2	
V2 Fractions, percentages	● ● ● ● ●	2	■ 14	
V3 Combining ratios	● ● ● ● ●	0	■ 18	
V4 Before and after	● ● ● ● ●	1	■ 1	
V5 Ratio with algebra	● ● ● ● ●	0	■ 3	

WHAT THIS TELLS YOU These are Question 1, 2 or 6 on Paper 1 — 3 or 4 marks with “write down all the stages”. They are easy marks if you label every step.

BEFORE YOU START

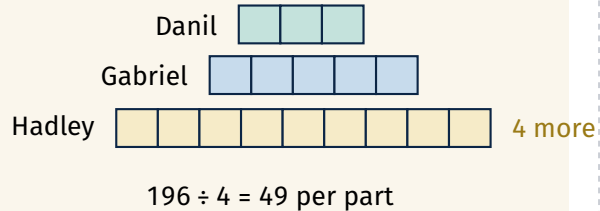
The big idea



Turn the words into parts.

Every ratio story gives you one fact in money, grams or people. Match that fact to its **number of parts**: a share, a **sum of shares** or a **difference**.

Divide to get **one part**. Everything else is multiplying.



Match the fact to its parts

A share

sugar =
10 parts
"She uses 120 g of sugar".

A sum

$2 + 6 = 8$ parts
"Asha and Julie together".

A difference

$9 - 5 = 4$ parts
"The difference between ...".

A rest

$1 - \frac{3}{10} = \frac{7}{10}$
"He changed the rest ...".



Label every step



CALCULATOR FIRST, THEN CHECK

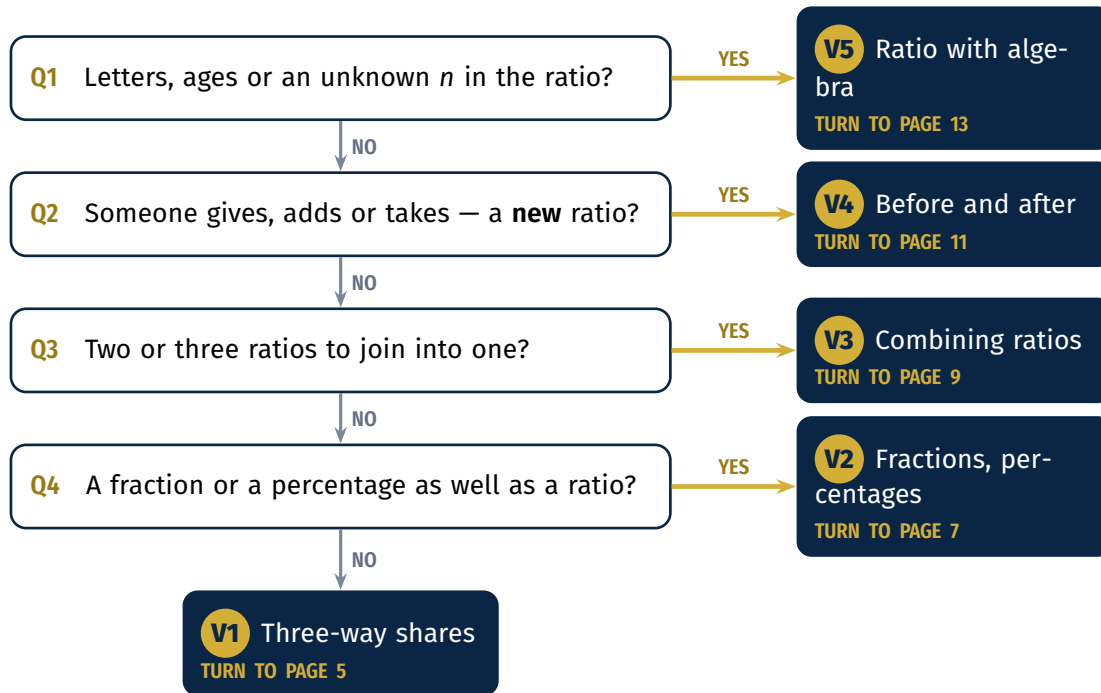
Check: the shares add to the total and divide back to the ratio. $147 : 245 : 441 = 3 : 5 : 9$ ✓

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of “How do I do it?” each line is.

ON EVERY VARIANT SPREAD



What is it?



How do I do it?



Worked example



Check yourself



When you see → you do



Watch out



Exam tip



Calculator check

V1

VARIANT 1 MOST COMMON IN THE EXAM

Three-Way Shares

The fact is a sum or a difference of two shares: divide by that many parts.

SEEN IN PAST PAPERS

● ● ● ● ● 2/5

PRACTICE QUESTIONS: 2

 Your notes

Came up in: Jan 2020 P1HR Q1 • Jan 2022 P1H Q2

? What is it?

In plain words

Three people share in 3 : 5 : 9. You are not given the total — you are given **two shares added** or **two shares subtracted**.

Add or subtract their **parts** in the same way, divide, then build the share you need.

IT LOOKS LIKE

$$\text{Asha} + \text{Julie} = \$36$$

$$\text{Hadley} - \text{Gabriel} = 196$$

$$B : A : J = 3 : 2 : 6$$

$$2 + 6 = 8 \text{ parts} = \$36$$

$$1 \text{ part} = \$4.50$$

SUM OF SHARES ↔ SUM OF PARTS

How do I do it?

1 Read

Which shares are added or subtracted?

2 Parts

Do the same to their ratio numbers.

3 One part

Divide the money by those parts.

4 Share

Multiply by the parts of the person asked for.

SAME SUM ON THE PARTS

Add shares → add parts; subtract → subtract.

V1 Three-Way Shares

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The total amount that Asha and Julie receive" → **Add their parts: $2 + 6$**

"The difference between ... Gabriel and Hadley" → **Subtract their parts: $9 - 5$**

"Write down all the stages in your working" → **Parts line, one-part line, answer line**

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 1H • Q2 • 3 marks

Danil, Gabriel and Hadley share some money in the ratios $3 : 5 : 9$. The difference between Gabriel's and Hadley's amounts is 196 euros. Work out Danil's amount.

STEP 1 Gabriel 5, Hadley 9 ◀ the two shares

STEP 2 $9 - 5 = 4$ parts = 196 ◀ difference of parts

STEP 3 1 part = $196 \div 4 = 49$ ◀ one part

STEP 4 Danil = $3 \times 49 = 147$ ◀ his share

ANSWER 147 euros✓ $441 - 245 = 196$ ✓**WATCH OUT**✗ $196 \div 17$ ✓ $196 \div (9 - 5)$

196 is a difference, not the total — use the difference of the parts.

EXAM TIP

Write the parts above the names:

$$\begin{array}{ccc} 3 & 5 & 9 \\ \text{D} & \text{G} & \text{H} \end{array} \text{ — then you can't mix them up.}$$
Check yourself

calculator first • answers on page 15

1.1 Jan 2020 P1HR

Brendon, Asha, Julie share in $3 : 2 : 6$. Asha and Julie receive \$36 in total. Brendon gets?

Answer:

1.2 Practice • Hard

£360: Abby : Ben = $2 : 7$; Chloe and Denesh each get 1.5 times Abby. Ben gets?

Answer:

1.3 Dr Eslam

A, B, C share in $2 : 3 : 7$. C gets £40 more than A and B together. Find the total.

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

Ratio with Fractions & Percentages

Take the fraction first, find the rest, then share the rest in the ratio.

SEEN IN PAST PAPERS

● ● ● ● ● **2 / 5**

PRACTICE QUESTIONS: 14



Your notes

Came up in: Jan 2021 P1H Q1 • Nov 2025 P1H Q6

? What is it?

In plain words

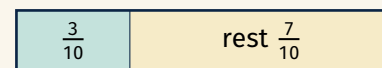
A fraction or percentage takes **one piece** out; the **rest** is shared in a ratio. Or: share first, then take a fraction of **each share**.

Read the order carefully and do it in that order.

IT LOOKS LIKE

 $\frac{3}{10}$ to euros, rest 9 : 5 : 2

30% female



share 9 : 5 : 2

euros out first, then the ratio

How do I do it?

1 Order

What happens first: fraction or ratio?

2 Fraction

Fraction/percentage of the right amount.

3 Rest

Subtract to get what is shared.

4 Share

One part, then the share(s) asked for.

FRACTION FIRST, THEN THE REST

In the order the story tells it.

V2 Ratio with Fractions & Percentages

VARIANT 2 · SEE IT DONE, THEN TRY

 Your notes
WHEN YOU SEE → YOU DO

"He changed $\frac{3}{10}$... the rest in the ratios 9 : 5 : 2" → **Rest = $\frac{7}{10}$ of the total, then share**

"Eli gives $\frac{3}{11}$ of his share" → **Share first, then the fraction of his share**

"30% are female; 1295 are male" → **70% = 1295: find 1%**

WORKED EXAMPLE PAST PAPER

January 2021 · Paper 1H · Q1 · 4 marks

Pieter changed 20 160 rand: $\frac{3}{10}$ into euros, the rest into dollars, rupees and francs in the ratios 9 : 5 : 2. How many more rand into dollars than into francs?

STEP 1 $\frac{3}{10} \times 20\,160 = 6048$ ◀ euros first

STEP 2 $20\,160 - 6048 = 14\,112$ ◀ the rest

STEP 3 $14\,112 \div 16 = 882$ ◀ one part (9 + 5 + 2)

STEP 4 $(9 - 2) \times 882 = 6174$ ◀ dollars - francs

ANSWER 6174 rand✓ $7938 - 1764 = 6174$ ✓**WATCH OUT**✗ $20\,160 \div 16$ ✓ $14\,112 \div 16$

Only the rest is shared in 9 : 5 : 2.

EXAM TIP

The question asks "how many more" — give the difference, not the dollars.

Check yourself

calculator first · answers on page 15

2.1

Nov 2025 P1H

Eli and Peta share \$275 in 2 : 3. Eli gives $\frac{3}{11}$ of his share, Peta 0.32 of hers to charity. Total given?

Answer:

2.2

Practice · Medium

30% of people at a concert are female; 1295 are male. How many are female?

Answer:

2.3

Dr Eslam

£600: $\frac{1}{4}$ is saved, the rest shared by Ali and Ben in 2 : 7. Ben gets?

Answer:



VARIANT 3

VARIANT 3

Combining Ratios

Match the shared quantity (LCM), join into one ratio, then use the total.

SEEN IN PAST PAPERS

● ● ● ● ● 0 / 5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 18



Your notes

What is it?

In plain words

Two ratios share a middle group:

men : women = 4 : 5, women :
children = 10 : 7.

Make women the **same number** in both (LCM 10): 8 : 10 : 7. Now it is **one ratio** — share the total.

IT LOOKS LIKE

4 : 5 and 10 : 7

“3 times as many red as green”

$$4 : 5 (\times 2) \rightarrow 8 : 10$$

$$10 : 7 \rightarrow 10 : 7$$

8 : 10 : 7 (25 parts)

MATCH THE MIDDLE

How do I do it?

1 Middle

Find the group in both ratios.

2 LCM

Scale so it is equal.

3 Join

One ratio; add the parts.

4 Use

Total $\div$ parts, or a fraction/percentage of the total.

MATCH THE MIDDLE “3 times as many” means 3 : 1.

V3 Combining Ratios

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"men : women = 4 : 5 ... women : children = 10 : 7"

→ **Match women (LCM 10)**

"three times as many red pens as green pens"

→ **red : green = 3 : 1**

"What percentage / fraction are blue?" → **Blue parts ÷ total parts**

WORKED EXAMPLE **PRACTICE**

Practice • Medium • Q12 • 4 marks

250 men, women and children on a train. Men : women = 4 : 5. Women : children = 10 : 7. How many men?

STEP 1 4 : 5 = 8 : 10

◀ women to 10

STEP 2 men : women : children = 8 : 10 : 7

◀ join

STEP 3 250 ÷ 25 = 10

◀ one part

STEP 4 8 × 10 = 80

◀ men

ANSWER 80 men

✓ 80 + 100 + 70 = 250 ✓

WATCH OUT

✗ 4 : 5 : 7

✓ 8 : 10 : 7

Women can't be 5 and 10 — match first.

EXAM TIP

A fraction or percentage of the whole:
use the joined ratio, $\frac{\text{parts}}{\text{total parts}}$.

Check yourself

calculator first • answers on page 15

3.1 Practice • Medium

3 times as many red pens as green; 2 times as many green as blue. Red : green : blue?

Answer:

3.2 Practice • Medium

Red : blue = 5 : 9; blue : green = 1 : 4. What percentage are blue?

Answer:

3.3 Practice • Medium

Houses : flats = 7 : 4; flats : bungalows = 8 : 5. 50 bungalows. How many houses?

Answer:

V4

VARIANT 4

Before & After

Work out the amounts before, make the change, then simplify the new ratio.

SEEN IN PAST PAPERS

● ● ● ● ● 1 / 5

PRACTICE QUESTIONS: 1



Your notes

Came up in: Jun 2025 P2H Q6

? What is it?**In plain words**

Something **changes**: money is given, counters are added. A ratio is about **one moment**, so find the real amounts **before**, apply the change, and write the **new** ratio.

If the change is unknown, use k for one part (Variant 5).

before	→	after
$60 : 80 : 100$		$80 : 70 : 90$

SAM, TIA -10 • PAU +20

 $= 8 : 7 : 9$ **IT LOOKS LIKE**

share £240, then each gives £10

5 red added

How do I do it?**1 Before**

Real amounts from the first ratio.

2 Change

Add/subtract to each person.

3 After

Write the new amounts as a ratio.

4 Simplify

Divide by the HCF.

REAL AMOUNTS, THEN CHANGE Never add to the ratio numbers.

V4 Before & After

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Sam and Tia each give £10 of their share to Pau" → **Pau +20, Sam -10, Tia -10**

"5 red counters are added" → **Real red count first, then +5**

"Give your answer in its simplest form" → **Divide by the HCF at the end**

WORKED EXAMPLE PAST PAPER

June 2025 · Paper 2H · Q6 · 4 marks

Pau, Sam and Tia share £240 in the ratios 3 : 4 : 5. Sam and Tia each give £10 of their share to Pau. Find the new ratio in its simplest form.

STEP 1 $240 \div 12 = 20 : 60, 80, 100$ ◀ before

STEP 2 Pau 60 + 20, Sam 80 - 10, Tia 100 - 10 ◀ change

STEP 3 80 : 70 : 90 ◀ after

STEP 4 8 : 7 : 9 ◀ $\div 10$

ANSWER 8 : 7 : 9✓ $80 + 70 + 90 = 240$ still ✓**WATCH OUT**

✗ $3 + 2 : 4 - 1 : 5 - 1$

✓ $60 + 20 : 80 - 10 : 100 - 10$

£10 is money, not a part — change the real amounts.

EXAM TIP

The total stays £240 when money only moves between people — a free check.

Check yourself

calculator first · answers on page 15

4.1

Dr Eslam

Red : blue = 2 : 5, 35 blue. 5 red are added. New red : blue?

Answer:

4.2

Dr Eslam

Amy : Ben = 5 : 3. Amy gives Ben £12; now they have the same. How much did Amy have?

Answer:

4.3

Dr Eslam

Boys : girls = 4 : 5 in a club of 36. 6 boys join. New boys : girls?

Answer:

V5

VARIANT 5

VARIANT 3

Write every amount with one letter, form an equation, solve, then build the ratio.

SEEN IN PAST PAPERS

● ● ● ● ● 0 / 5

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 3



Your notes

? What is it?

In plain words

When the facts are “7 years older”, “twice as old”, “7 : n ”, use **one letter**. Write every amount in terms of it, use the total to form an **equation**, solve, then write the ratio and simplify.

Jay = J

$$\text{Kiaria} = J + 7$$
$$\text{Martha} = 2(J + 7)$$

IT LOOKS LIKE

Kiaria is 7 years older than Jay

 $7 : n$
$$4J + 21 = 77$$

How do I do it?

1 Letter

Let the smallest amount be x .

2 Express

Every other amount in terms of x .

3 Equation

Use the total (or a ratio as a fraction); solve.

4 Ratio

Write the amounts; simplify.

ONE LETTER FOR EVERYONE Then it's Linear Equations (Chapter 2).

V5 Ratio with Algebra

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“7 years older than”, “twice as old as” → $J + 7, 2(J + 7)$

“Joe gets 35% ... ratio $7 : n$ ” → $\frac{7}{7+n} = 0.35$

“Find the ratio ... in its simplest form” → **Solve first, then divide by the HCF**

WORKED EXAMPLE **PRACTICE**

Practice • Hard • Q3 • 4 marks

Kiaria is 7 years older than Jay. Martha is twice as old as Kiaria. The sum of their ages is 77. Find Jay : Kiaria : Martha.

STEP 1 $J, J + 7, 2J + 14$ ◀ one letter

STEP 3 $4J + 21 = 77 \Rightarrow J = 14$ ◀ equation, solve

STEP 4 $14 : 21 : 42 = 2 : 3 : 6$ ◀ ratio, $\div 7$

ANSWER $2 : 3 : 6$ $\checkmark 14 + 21 + 42 = 77 \checkmark$ **WATCH OUT**✗ Martha = $2J + 7$ ✓ Martha = $2(J + 7)$

Twice Kiaria, so the bracket matters.

EXAM TIP

Answer the ratio, not just J — the last mark is the simplified ratio.

Check yourself

calculator first • answers on page 15

5.1 Practice • Medium

Joe and Kyle share money in the ratio $7 : n$. Joe gets 35%. Find n .

Answer:

5.2 Dr Eslam

$x : y = 2 : 5$. Find $(3x + y) : (x + 2y)$.

Answer:

5.3 Dr Eslam

$(a + b) : (a - b) = 5 : 2$. Find $a : b$.

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 \$13.50**1.2** £168**1.3** £240**2.1** \$82.80**2.2** 555**2.3** £350**3.1** 6 : 2 : 1**3.2** 18%**3.3** 140**4.1** 19 : 35**4.2** £60**4.3** 11 : 10**5.1** $n = 13$ **5.2** 11 : 12**5.3** 7 : 3

THE WHOLE TOPIC ON ONE PAGE

Ratio problems cheat sheet



Your notes
V1 Three-way 2/5

$$196 \div (9 - 5) = 49$$

Sum or difference of shares $\leftrightarrow$ of parts.**V2 Fractions, % 2/5**

$$\text{rest} = \frac{7}{10} \rightarrow 9 : 5 : 2$$

Fraction first, then share the rest.

V3 Combining

$$4 : 5, 10 : 7 \rightarrow 8 : 10 : 7$$

Match the middle, join.

V4 Before & after 1/5

$$60 : 80 : 100 \rightarrow 8 : 7 : 9$$

Real amounts, change, simplify.

V5 Algebra

$$J, J + 7, 2(J + 7)$$

One letter, equation, ratio.

**TOP 3 LOST MARKS**

1. dividing by the total parts for a difference $\times$
2. sharing everything, not the rest $\times$
3. changing the ratio numbers $\times$

BOTTOM LINE Fact $\rightarrow$ parts · one part first · match the middle · real amounts before you change.